

Lipid Metabolism

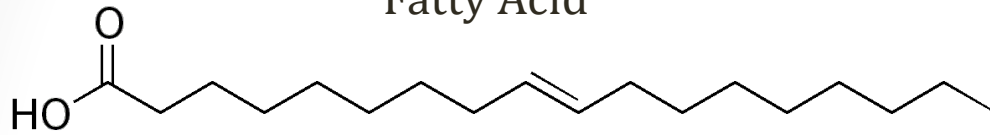
Jason Ryan, MD, MPH

Lipids

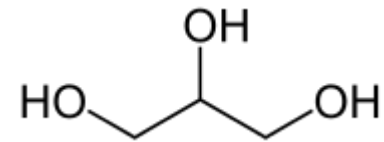
- Mostly carbon and hydrogen
- Not soluble in water
- Many types:
 - Fatty acids
 - Triglycerides
 - Cholesterol
 - Phospholipids
 - Steroids
 - Glycolipids

Lipids

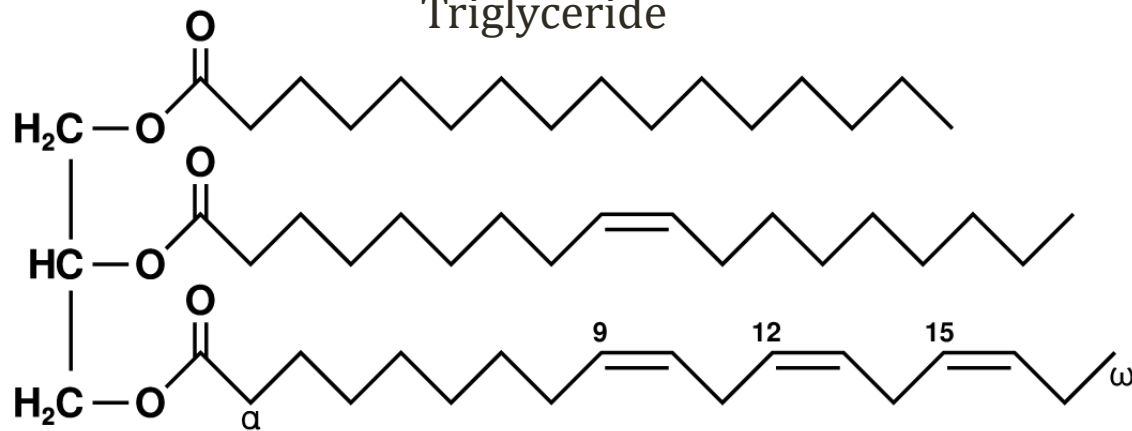
Fatty Acid



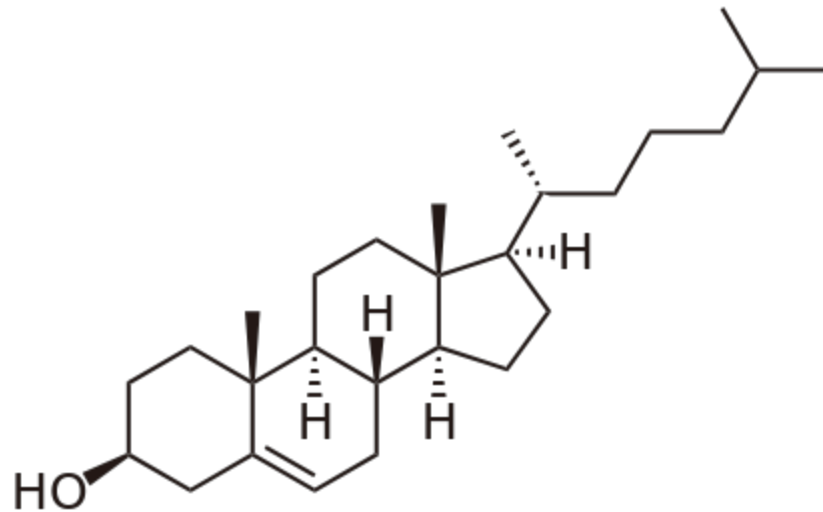
Glycerol



Triglyceride



Lipids

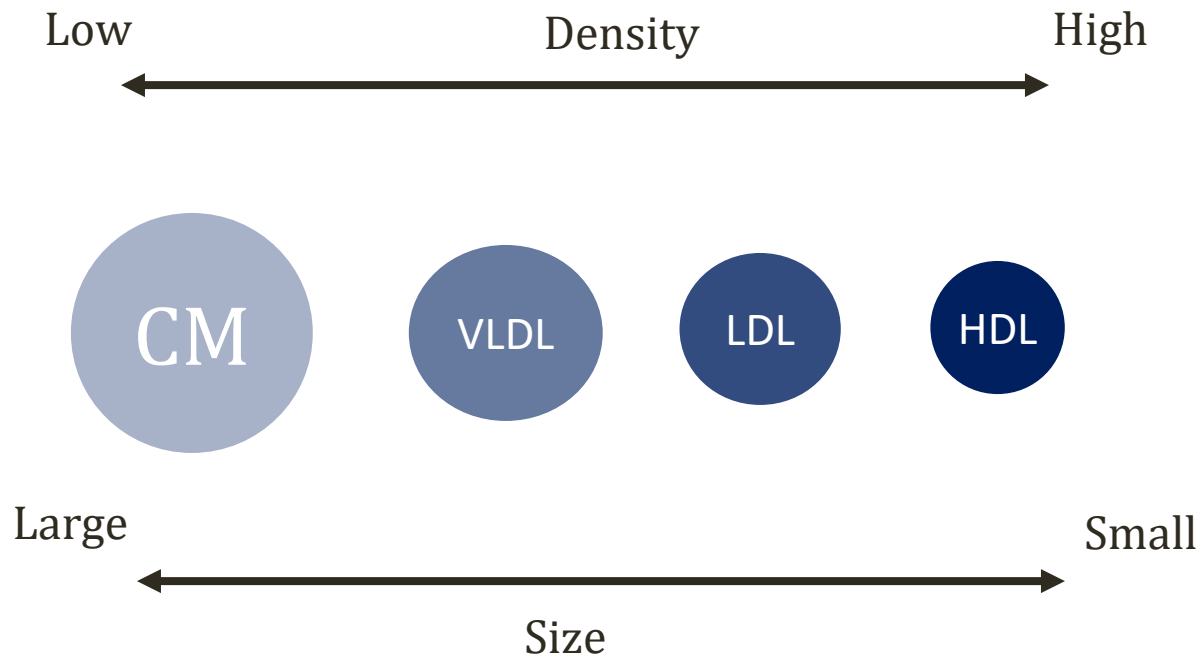


Cholesterol

Lipoproteins

Particles of lipids and proteins

- Chylomicrons
- Very low-density lipoprotein (VLDL)
- Intermediate-density lipoprotein (IDL)
- Low density lipoproteins (LDL)
- High-density lipoprotein (HDL)

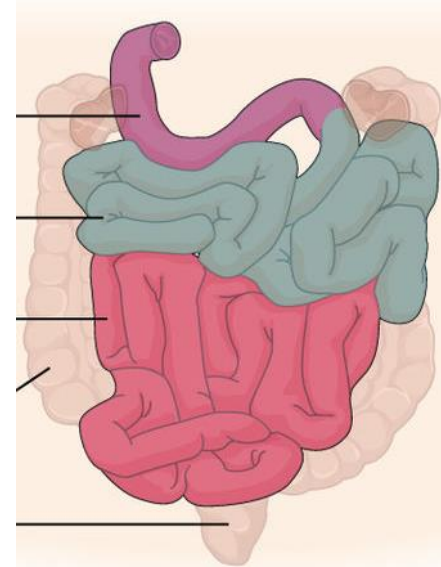


Apolipoproteins

- Proteins that bind lipids
- Found in lipoproteins
- Various functions:
 - Surface receptors
 - Co-factors for enzymes

Absorption of Fatty Acids

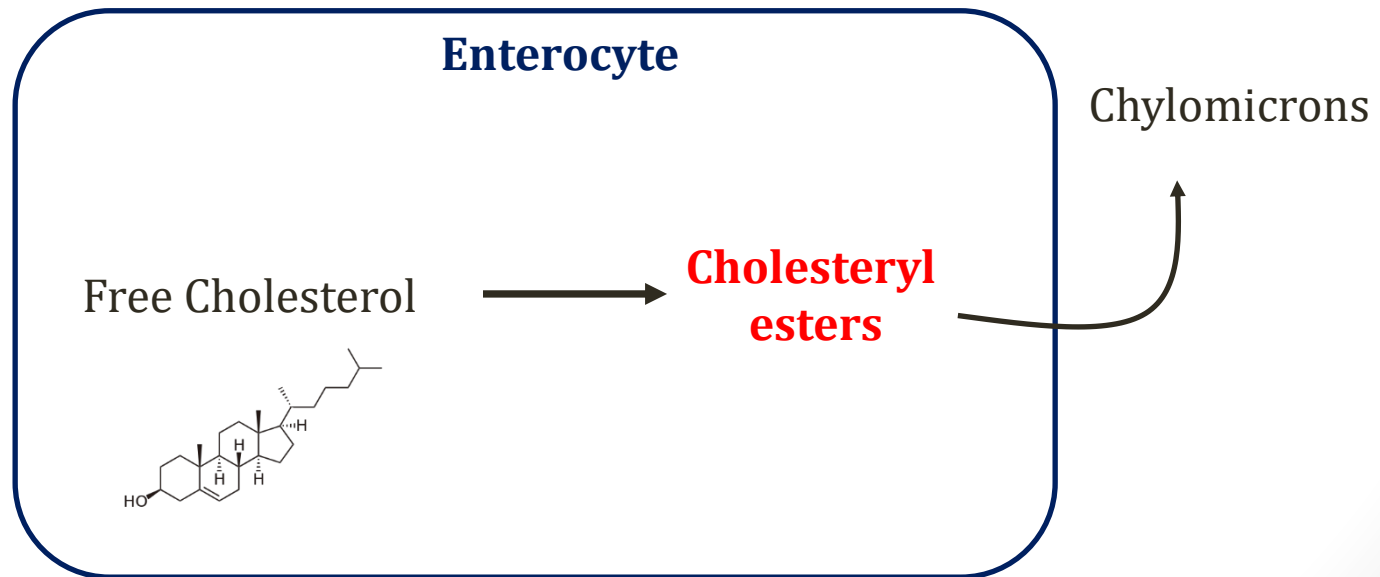
- Fatty acids → Triglycerides
- Packaged into **chylomicrons** by intestinal cells
- To lymph → blood stream



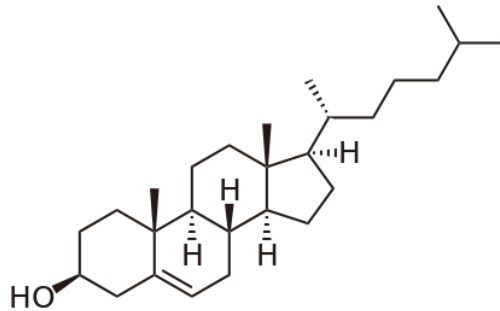
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Absorption of Cholesterol

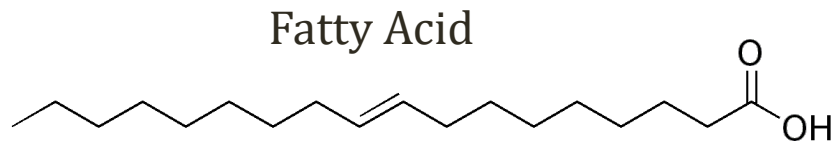
- **Cholesteryl esters** formed in enterocytes
- Acyl-CoA cholesterol acyltransferase (**ACAT**)
- Packaged into chylomicrons by intestinal cells
- To lymph → blood stream



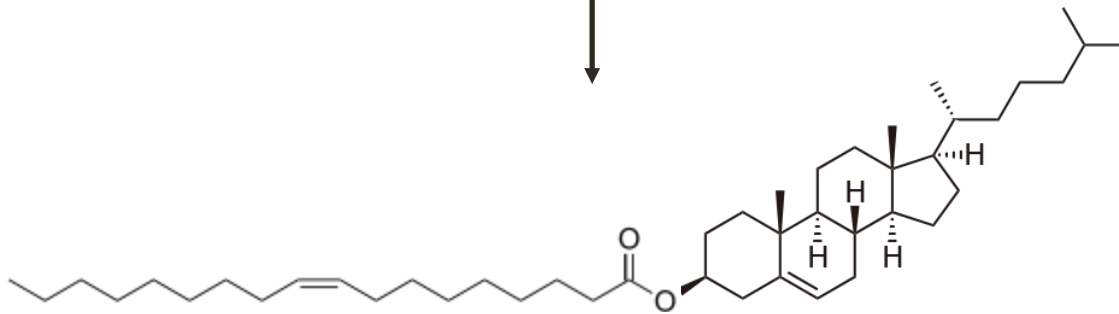
Cholesteryl Esters



Cholesterol

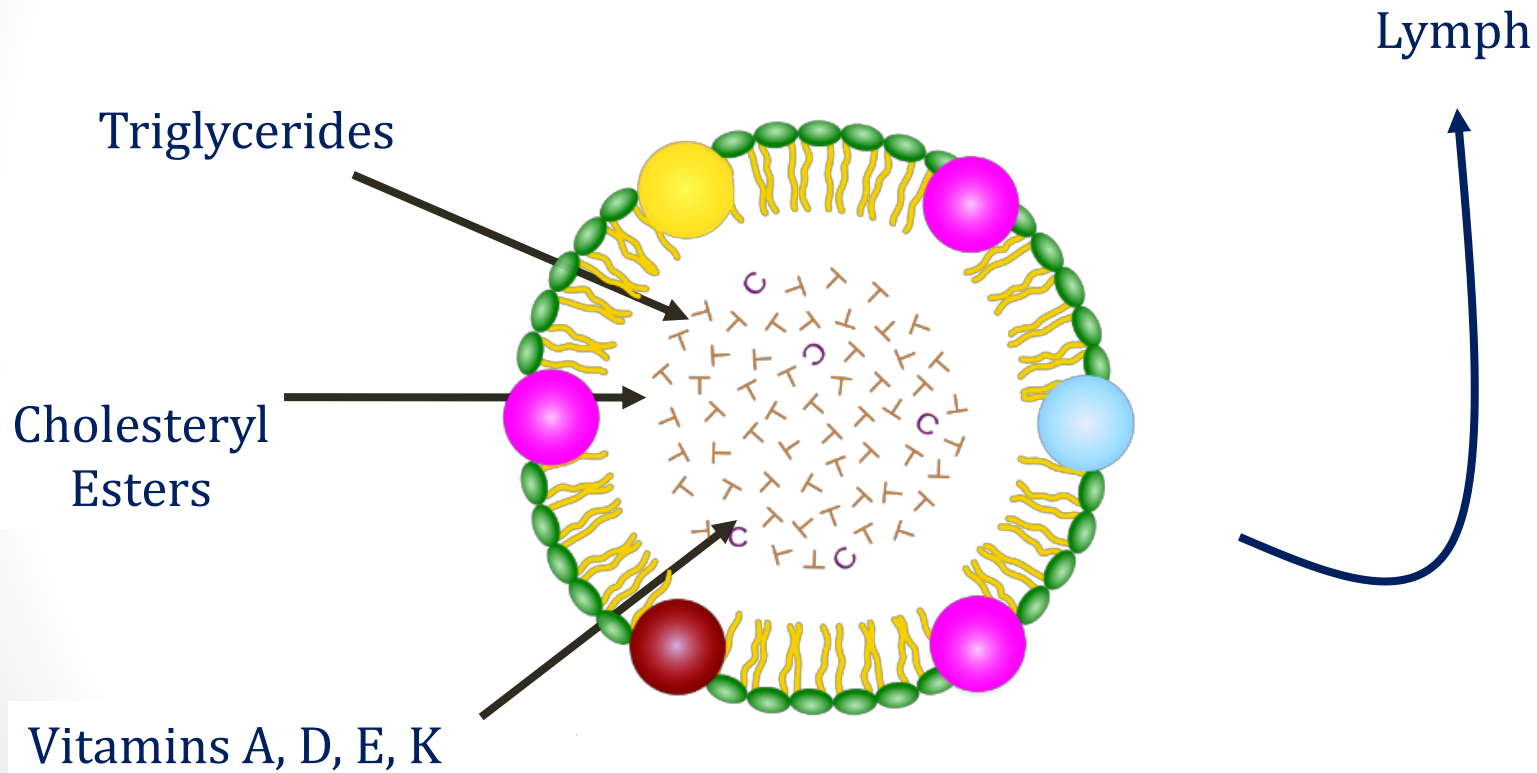


ACAT



Cholesteryl Ester

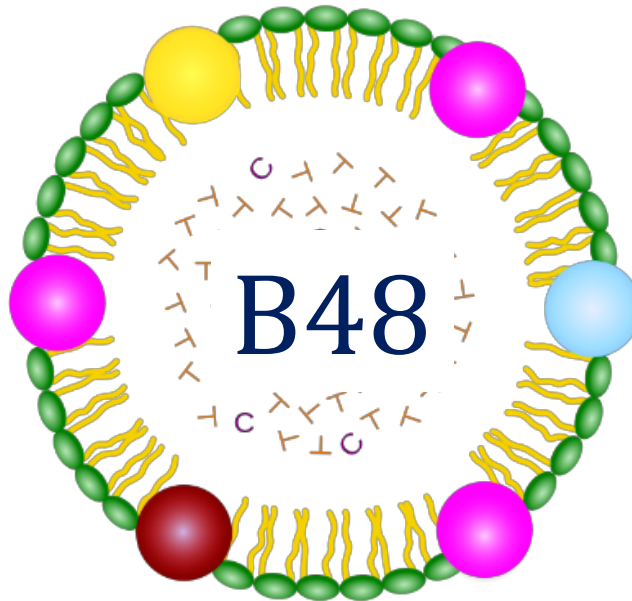
Chylomicrons



Open Stax College/Wikipedia

Apolipoprotein B48

- Found only on chylomicrons
- Contains 48% of apo-B protein
- Required for **secretion from enterocytes**



Open Stax College/Wikipedia

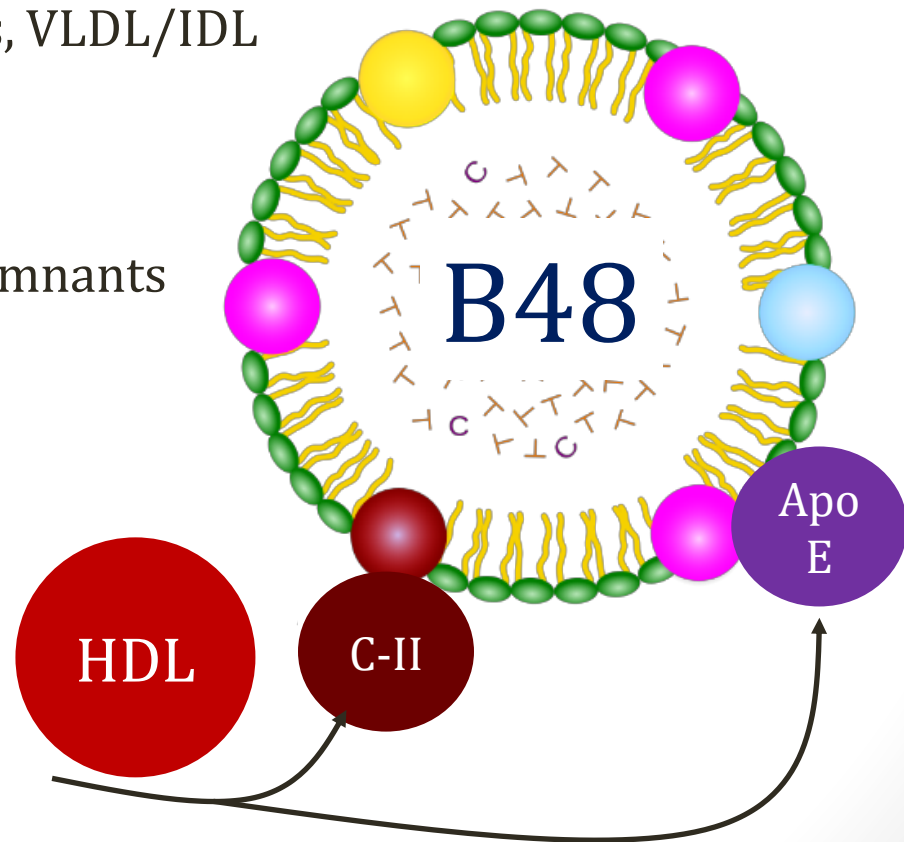
Lipoprotein Lipase

LPL

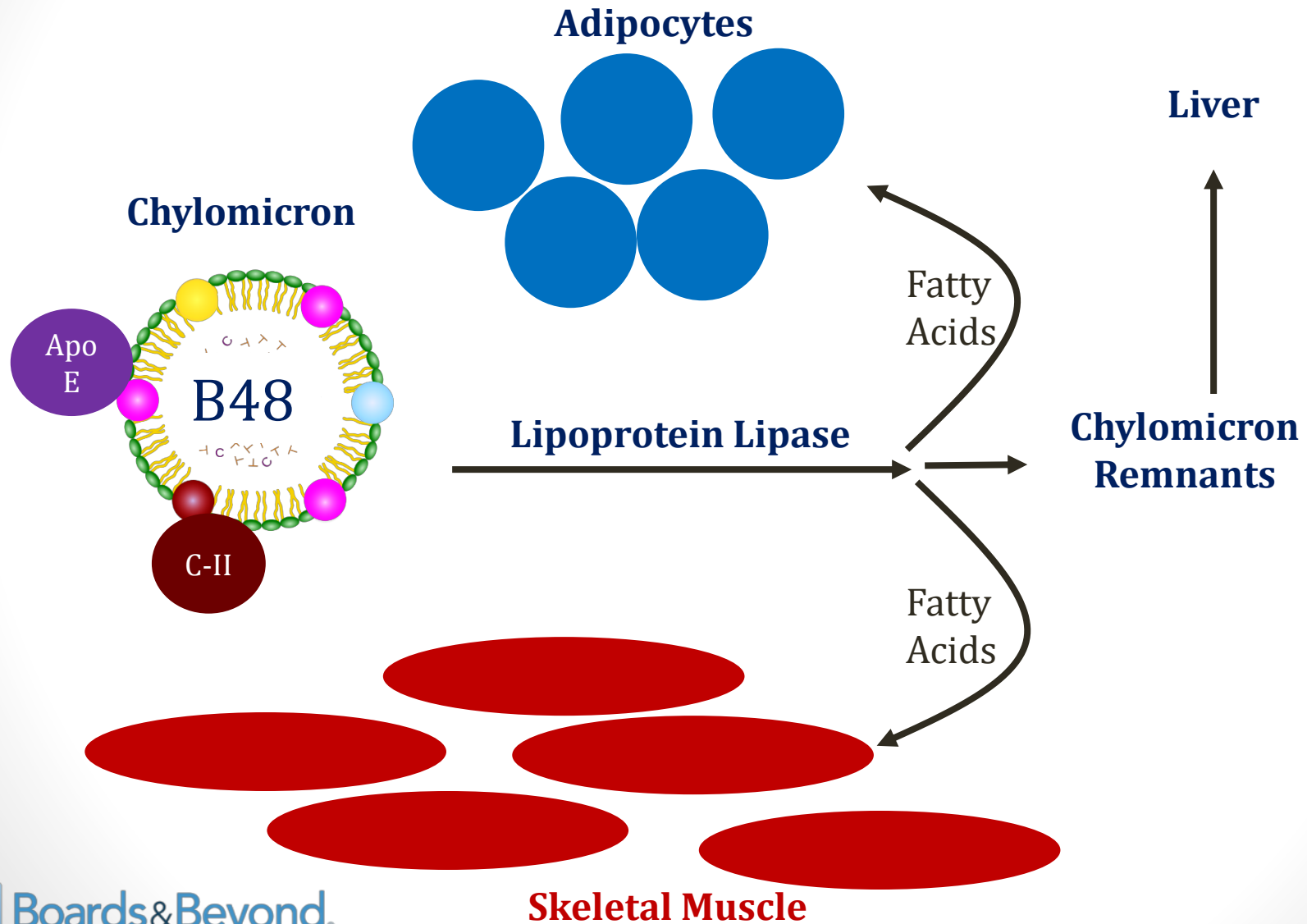
- **Extracellular** enzyme
- Anchored to **capillary walls**
- Mostly found in adipose tissue, muscle, and heart
 - Not in liver → liver has hepatic lipase
- Converts triglycerides → fatty acids (and glycerol)
- Fatty acids used for storage (adipose) or fuel
- Requires **apo C-II** for activation

Other Apolipoproteins

- **C-II**
 - Co-factor for lipoprotein lipase
 - Carried by: Chylomicrons, VLDL/IDL
- **Apo E**
 - Binds to liver receptors
 - Required for uptake of remnants
- Both from **HDL**

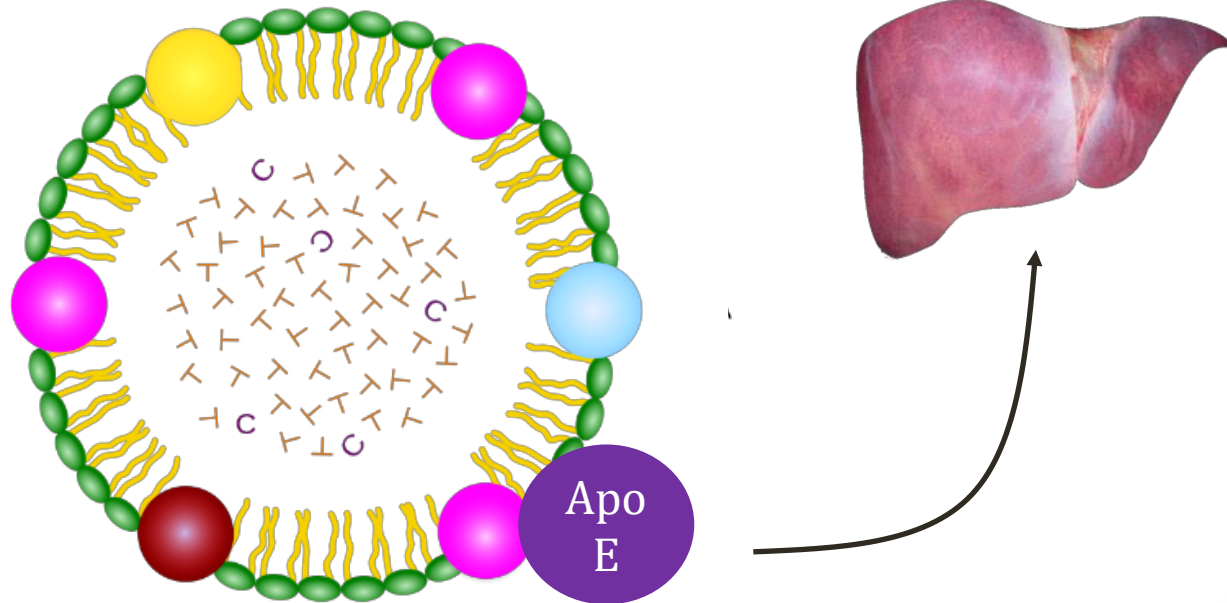


Chylomicrons



Chylomicron Remnants

- Apo-E receptors on **liver**
- Take up remnants via receptor-mediated endocytosis
- Usually only present after meals (clear 1-5hrs)
- Milky appearance



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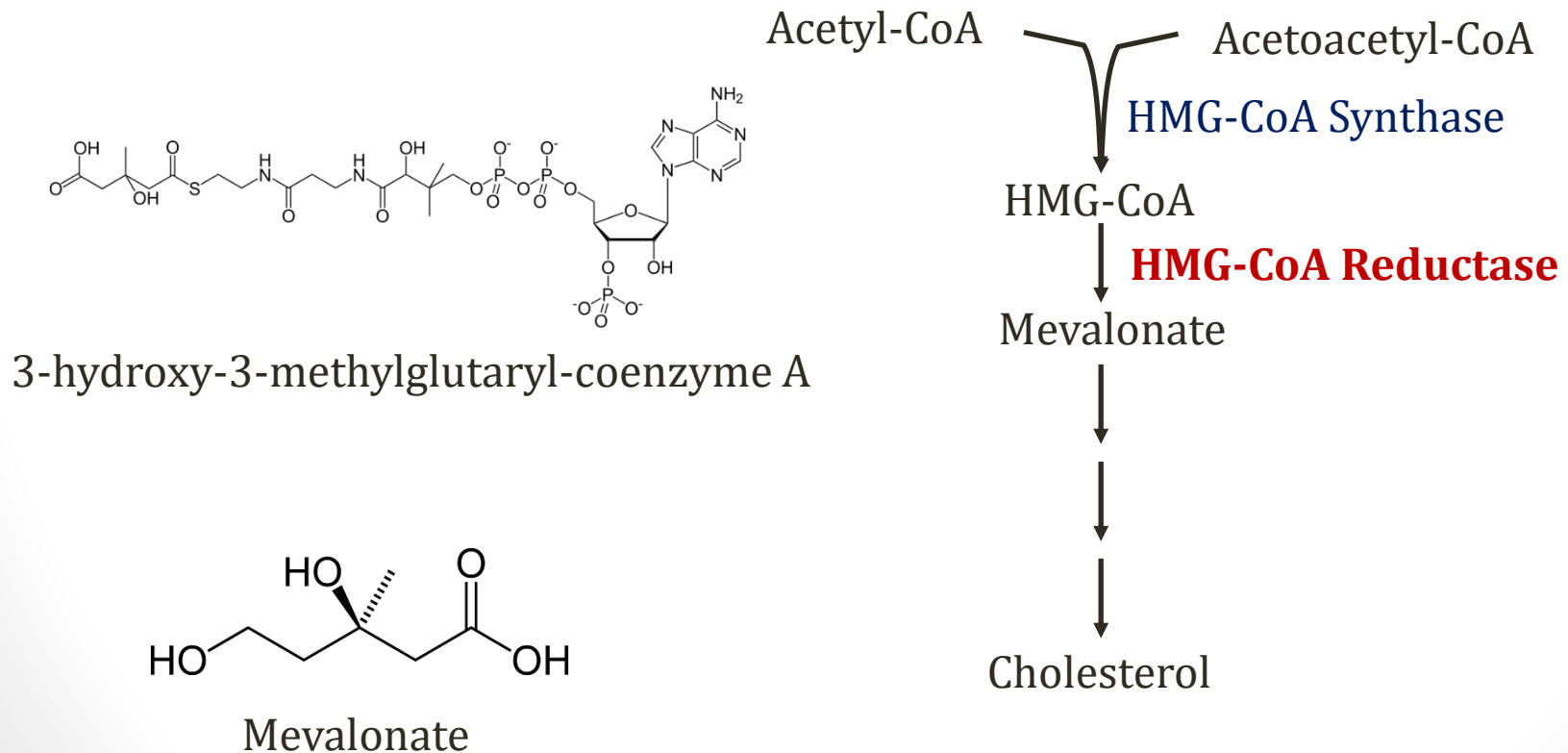
Chylomicrons

Summary

- Secreted from enterocytes with Apo48
- Pick up Apo C-II and ApoE from HDL
- Carry triglycerides and cholesteryl esters
- Deliver triglycerides to cells
 - Lipoprotein lipase stimulates (C-II co-factor) breakdown
- Return to liver as chylomicron remnants
 - ApoE receptors on liver

Cholesterol Synthesis

- Only the liver can synthesize cholesterol



Lipid Transport

- Liver secretes two main lipoproteins:
 - VLDL
 - HDL

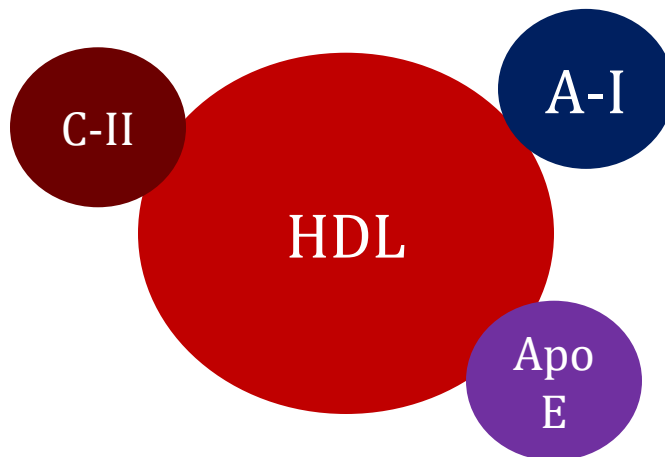


VLDL
HDL

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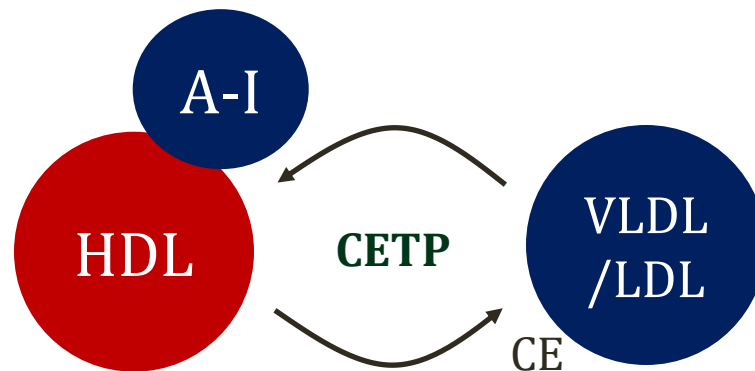
HDL

- **Scavenger** lipoprotein
 - Brings cholesterol back to liver
 - “Reverse transport”
- Secreted as small “nascent” HDL particle
- Key apolipoproteins: A-I, C-II, ApoE



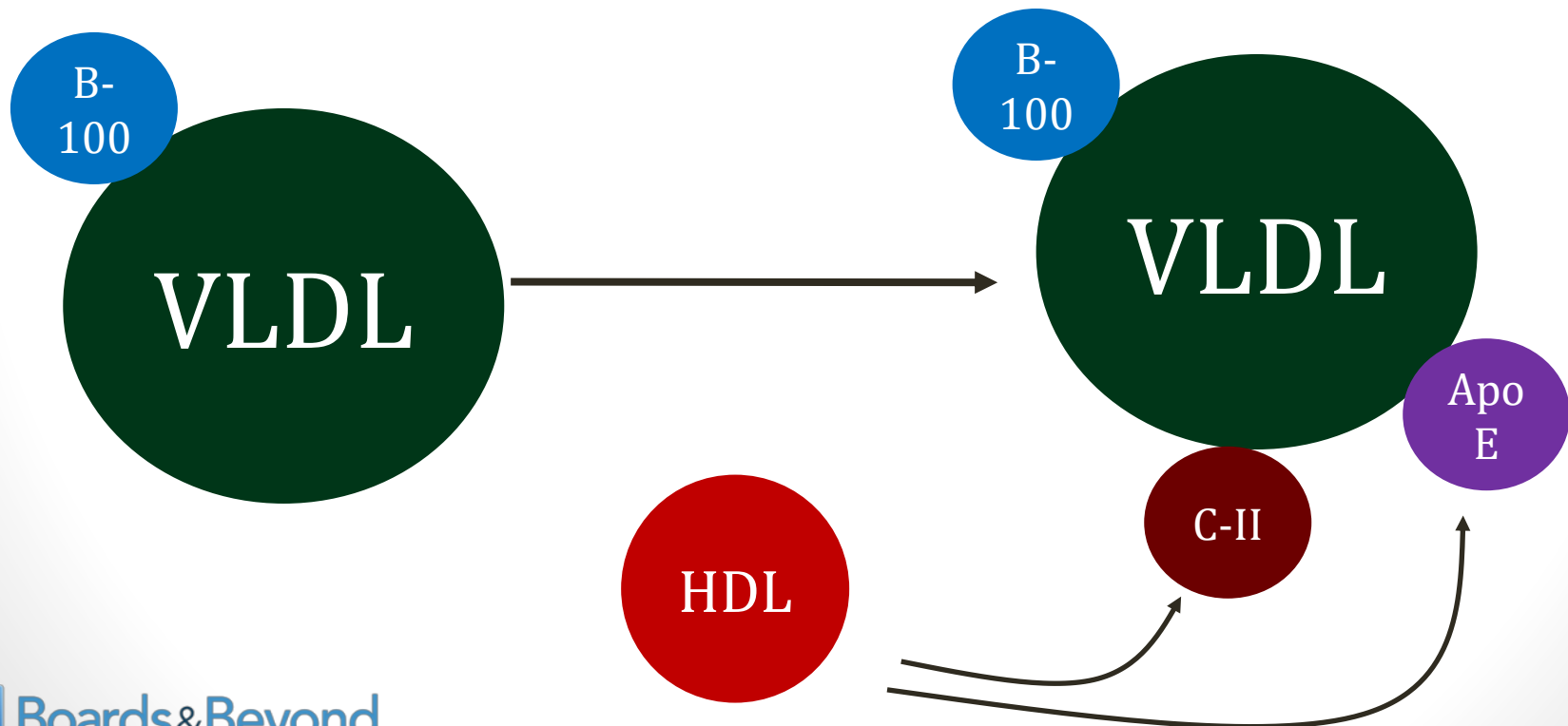
HDL

- Lecithin-cholesterol acyl transferase (**LCAT**)
 - Esterifies cholesterol in HDL; packs **esters** densely in core
 - Activated by **A-I**
- Cholesteryl ester transfer protein (CETP)
 - Exchanges esters (HDL) for triglycerides (VLDL)
- Carries cholesterol back to liver



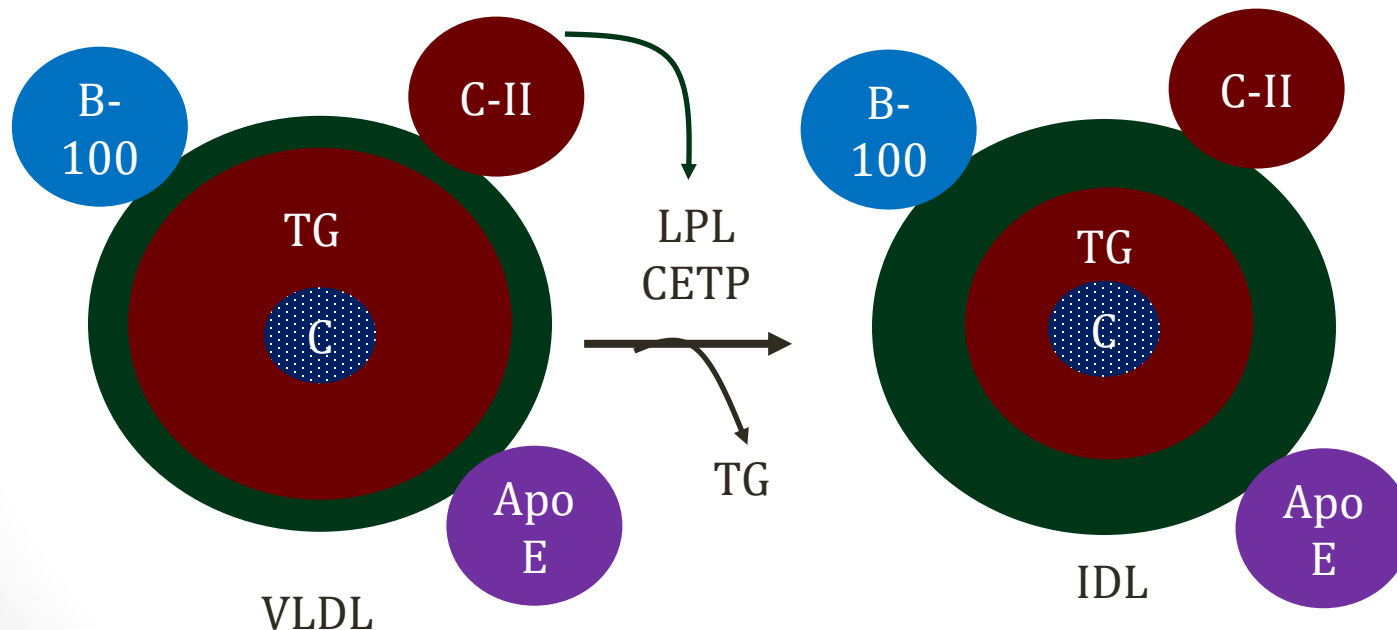
VLDL

- **Transport** lipoprotein
- Secreted by liver (nascent VLDL)
- Carries triglycerides, cholesterol to tissues



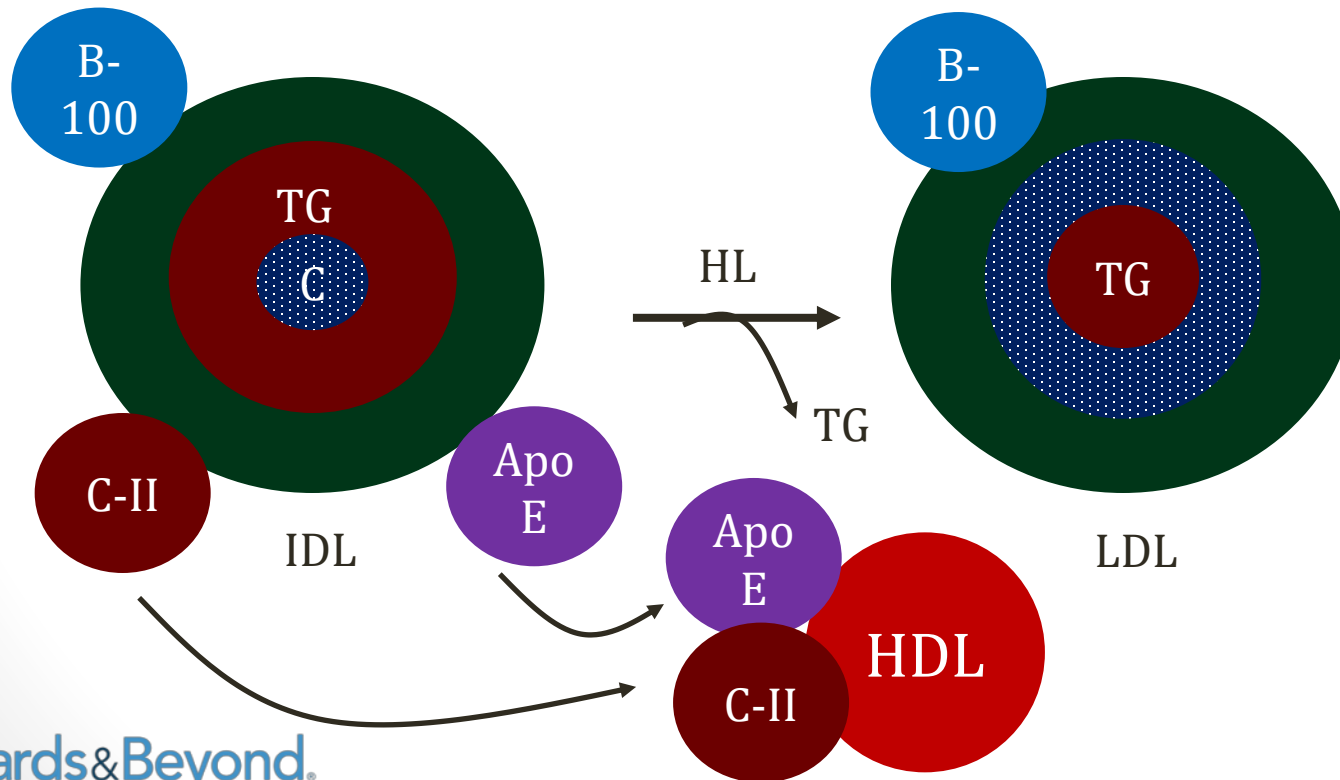
VLDL

- Changes during circulation
- #1: LPL removes triglycerides
- #2: CETP in HDL removes triglycerides from VLDL



IDL

- Formed from VLDL
- Hepatic lipase removes triglycerides
- HDL removes C-II and ApoE

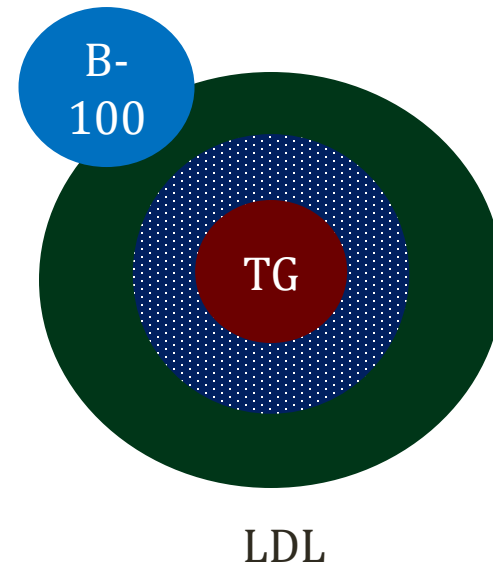


Hepatic Lipase

- Found in **liver** capillaries
- Similar function to LPL (releases fatty acids)
- Very important for IDL → LDL conversion
- Absence HL → absence IDL/LDL conversion

LDL

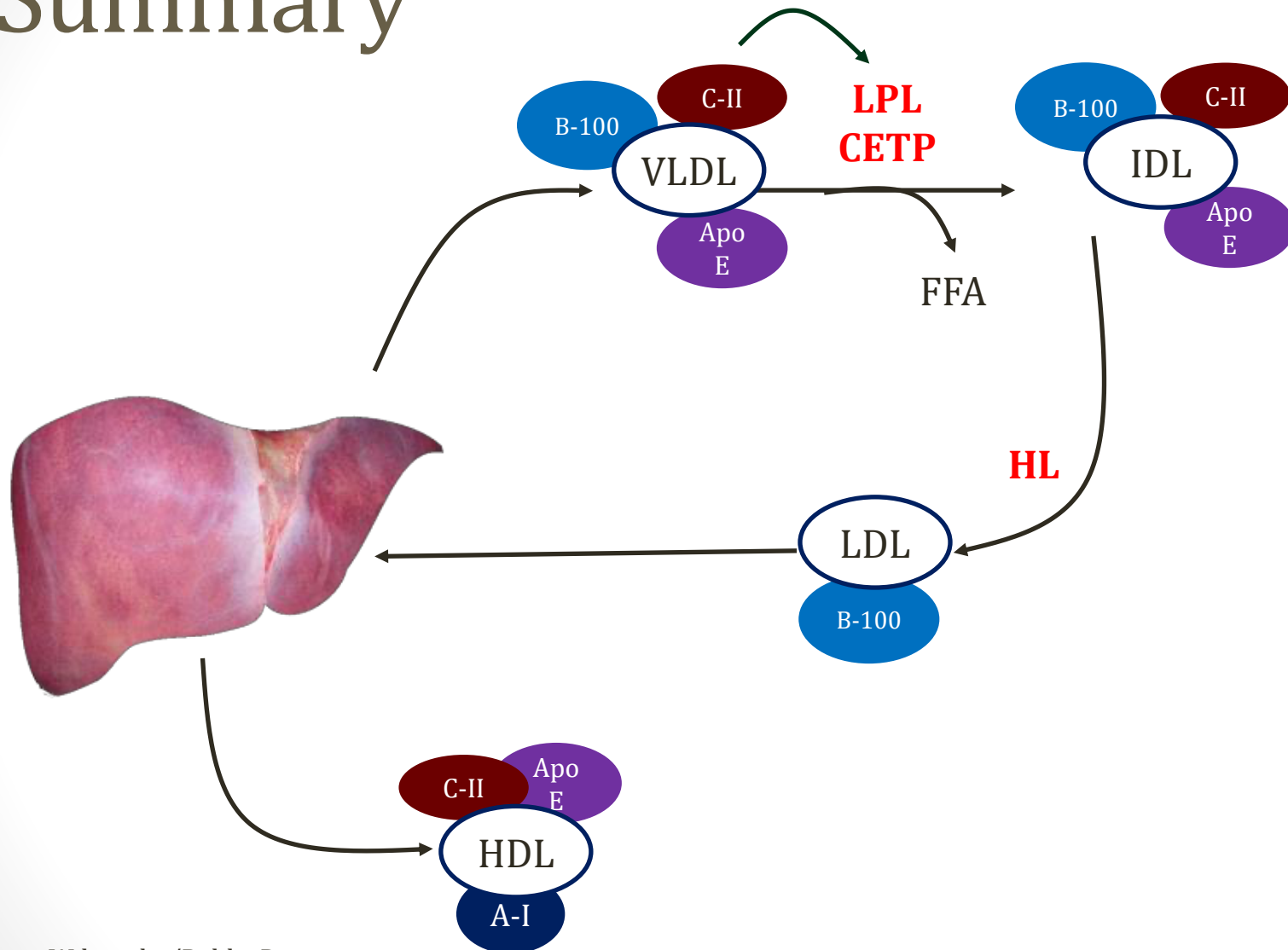
- Small amount of triglycerides
- High concentration of cholesterol/cholesteryl esters
- Transfers cholesterol to cells with **LDL receptor**
 - Receptor-mediated endocytosis
- LDL receptors recognize B100



Foam Cells

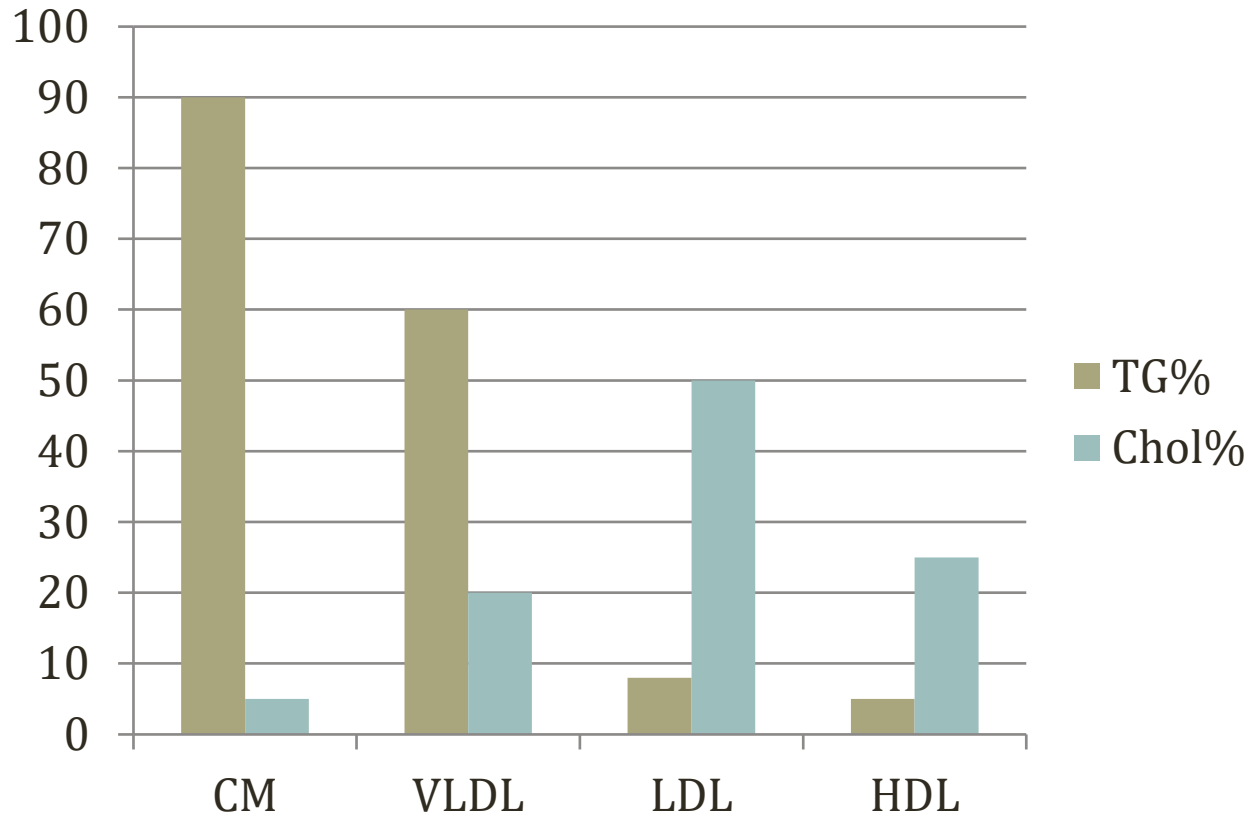
- **Macrophages** filled with cholesterol
- Found in atherosclerotic plaques
- Contain LDL receptors and LDL

Summary



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Lipoprotein Composition



Lipoprotein(a)

Lp(a)

- Modified form of LDL
- Contains large glycoprotein apolipoprotein(a)
- Elevated levels risk factor for cardiovascular disease
- Not routinely measured
- No proven therapy for high levels

Abetalipoproteinemia

- Autosomal recessive disorder
- Defect in MTP
 - Microsomal triglyceride transfer protein
- MTP forms/secretes lipoproteins with apo-B
 - Chylomicrons from intestine (B48)
 - VLDL from liver (B100)

Abetalipoproteinemia

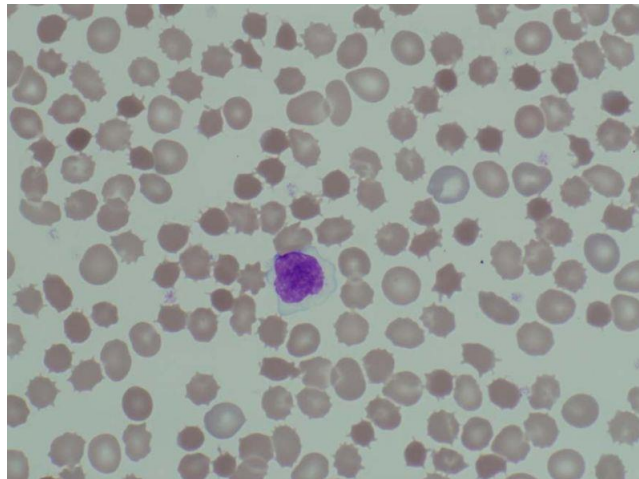
Clinical Features

- Presents in infancy
 - Steatorrhea
 - Abdominal distension
 - Failure to thrive
- Fat-soluble vitamin deficiencies
 - Especially vitamin E (ataxia, weakness, hemolysis)
 - Vitamin A (poor vision)
- Lipid accumulation in enterocytes on biopsy

Abetalipoproteinemia

Lab Findings

- Low or zero VLDL/ILDL/LDL
- Very low triglyceride and total cholesterol levels
- Low vitamin E levels
- Acanthocytosis
 - Abnormal RBC membrane lipids



Rola Zamel, Razi Khan,
Rebecca L Pollex and Robert A Hegele

Hyperlipidemia

Jason Ryan, MD, MPH

Lipid Measurements

- Total Cholesterol
- LDL-C
- HDL-C
- TG

Friedewald Formula

$$\text{LDL-C} = \text{Total Chol} - \text{HDL-C} - \frac{\text{TG}}{5}$$

Hyperlipidemia

- Elevated total cholesterol, LDL, or triglycerides
- Risk factor for coronary disease and stroke
- Modifiable – often related to lifestyle factors
 - Sedentary lifestyle
 - Saturated and trans-fatty acid foods
 - Lack of fiber

Secondary Hyperlipidemia

Selected Causes of Hyperlipidemia
Nephrotic syndrome (LDL) Alcoholism (TG) Pregnancy (TG>TC) Beta blockers (TG) HCTZ (TC, LDL, TG)

Signs of Hyperlipidemia

- Most patients have no signs/symptoms
- Physical findings occur in patients with severe ↑lipids
- Usually familial syndrome

Signs of Hyperlipidemia

- Xanthomas
 - Plaques of lipid-laden histiocytes
 - Appear as skin bumps or on eyelids
- Tendinous Xanthoma
 - Lipid deposits in tendons
 - Common in Achilles
- Corneal arcus
 - Lipid deposit in cornea
 - Seen on fundoscopy



Klaus D. Peter, Gummersbach, Germany



Min.neel/Wikipedia

Pancreatitis

- Elevated triglycerides (>1000) $\rightarrow$ acute pancreatitis
- Exact mechanism unclear
- May involve increased **chylomicrons** in plasma
 - Chylomicrons usually formed after meals and cleared
 - Always present when triglycerides $> 1000\text{mg/dL}$
 - May obstruct capillaries $\rightarrow$ ischemia
 - Vessel damage can expose triglycerides to pancreatic lipases
 - Triglycerides breakdown $\rightarrow$ **free fatty acids**
 - Acid $\rightarrow$ tissue injury $\rightarrow$ pancreatitis

Familial Dyslipidemias

- Type I – **Hyperchylomicronemia**
- Autosomal recessive
- ↑↑↑TG (>1000; milky plasma appearance)
- ↑↑↑ chylomicrons

Familial Dyslipidemias

- Type I – **Hyperchylomicronemia**
- Severe LPL dysfunction
 - LPL deficient
 - LPL co-factor deficient (apolipoprotein C-II)
- Recurrent pancreatitis
- Enlarged liver, xanthomas
- Treatment: Very low fat diet
 - Reports of normal lifespan
 - No apparent ↑risk atherosclerosis

Familial Dyslipidemias

- Type II - **Familial Hypercholesterolemia**
 - Autosomal **dominant**
 - Few or zero LDL receptors
 - **Very high LDL** (>300 heterozygote; >700 homozygote)
 - Tendon xanthomas, corneal arcus
 - **Severe atherosclerosis** (can have MI in 20s)

Familial Dyslipidemias

- Type III – **Familial Dysbetalipoproteinemia**
 - Apo-E2 subtype of Apo-E
 - Poorly cleared by liver
 - Accumulation of chylomicron remnants and VLDL
 - (collectively know as β -lipoproteins)
 - Elevated total cholesterol and triglycerides
 - Usually mild (TC>300 mg/dl)
 - Xanthomas
 - **Premature coronary disease**

ApoE and Alzheimer's

- ApoE2
 - Decreased risk of Alzheimer's
- ApoE4
 - Increased risk of Alzheimer's

Familial Dyslipidemias

- Type IV **Hypertriglyceridemia**
 - Autosomal **dominant**
 - VLDL overproduction or impaired catabolism
 - **↑TG (200-500)**
 - ↑VLDL
 - Associated with diabetes type II
 - Often diagnosed on routine screening bloodwork
 - Increased coronary risk/premature coronary disease

Lipid Drugs

Jason Ryan, MD, MPH

The “Cholesterol Panel”

“Lipid Panel”

- Total Cholesterol
- LDL-C
- HDL-C
- TG

Friedewald Formula

$$\text{LDL} = \text{Total Chol} - \text{HDL} - \frac{\text{TG}}{5}$$

LDL Cholesterol

- “Bad” cholesterol
- Associated with CV risk
- <100 mg/dl very good
- >200 mg/dl high
- Evidence that treating high levels reduces risk

HDL Cholesterol

- “Good” cholesterol
- Inversely associated with risk
- <45mg/dl low
- Little evidence that raising low levels reduces risk

Triglycerides

- Normal TG level <150mg/dl
- Levels > 1000 can cause **pancreatitis**
- Elevated TG levels modestly associated with CAD
- Little evidence that lowering high levels reduces risk

Pancreatitis

- Elevated triglycerides → acute pancreatitis
- Exact mechanism unclear
- May involve increased **chylomicrons** in plasma
 - Chylomicrons usually formed after meals and cleared
 - Always present when triglycerides > 1000mg/dL
 - May obstruct capillaries → ischemia
 - Vessel damage can expose triglycerides to pancreatic lipases
 - Triglycerides breakdown → free fatty acids
 - Acid → tissue injury → pancreatitis

Treating Hyperlipidemia

- Usually treat **elevated LDL-C with statins**
- Rarely treat elevated TG or low HDL-C
- Secondary prevention
 - Patients with coronary or vascular disease
 - Strong evidence that lipid lowering drugs benefit
- Primary prevention
 - Not all patients benefit the same
 - Benefit depends on risk of CV disease

Guidelines

Lipid Drug Therapy

- Old Cholesterol Guidelines set LDL-C goal
 - Diabetes or CAD: Goal LDL <100
 - 2 or more risk factors: Goal LDL <130
 - 0 or 1 risk factor: Goal LDL <160
- New guidelines require risk calculator
 - Treat patients if risk above limit (usually 5%/year)
 - No LDL goal
- **Statins 1st line** majority of hyperlipidemia patients

Treating TG or HDL

- Rarely treat for TG or HDL alone
- Many LDL drugs improve TG/HDL
- Few data showing a benefit of treatment

Treating TG or HDL

- Triglycerides
 - >500
 - High Non-HDL cholesterol (TC – HDL)
- Low HDL
 - Patients with established CAD

Lipid Lowering Drugs

- Statins
- Niacin
- Fibrates
- Absorption blockers
- Bile acid resins
- Omega-3 fatty acids

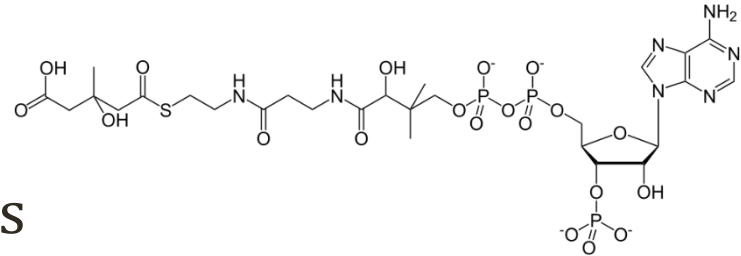
Diet/exercise/weight loss = GREAT way to reduce
cholesterol levels and CV risk

Statins

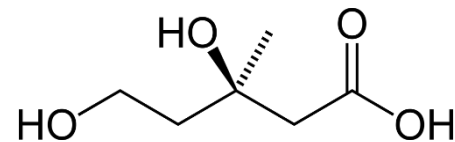
Lovastatin, Atorvastatin, Simvastatin

- Act on liver synthesis
- HMG-CoA reductase inhibitors
- ↓cholesterol synthesis in liver
- **↑LDL receptors in liver**
- Major effect: ↓ LDL decrease
- Some ↓TG, ↑HDL
- **Excellent outcomes data** (↓MI/Death)
- ↑LFTs

3-hydroxy-3-methylglutaryl-coenzyme A
HMG-CoA



**HMG-CoA
Reductase**



Mevalonate

Statin Muscle Problems

- Many muscle symptoms associated with statins
- Mechanism poorly understood
- Low levels of **coenzyme Q** in muscles
- Many patients take CoQ 10 supplements
- Theoretical benefit for muscle aches on statins

Statin Muscle Problems

- Myalgias
 - Weakness, soreness
 - Normal CK levels
- Myositis
 - Like myalgias, increased CK
- Rhabdomyolysis
 - Weakness, muscle pain, dark urine
 - CKs in 1000s
 - Acute renal failure → death
 - ↑risk with some drugs (cyclosporine, gemfibrozil)

Hydrophilic vs. Lipophilic

Statins

- Hydrophilic statins
 - Pravastatin, fluvastatin, **rosuvastatin**
 - May cause less myalgias
- Lipophilic statins
 - **Atorvastatin**, **simvastatin**, lovastatin

P450

- Statins metabolized by liver P450 system
- Interactions with other drugs
- Inhibitors increase ↑ risk LFTs/myalgias
 - i.e. grapefruit juice



Citrus_paradisi/Wikipedia

Niacin

- Complex, incompletely understood mechanism
- Overall effect: LDL ↓↓ **HDL ↑↑**
- Often used when HDL is low

Niacin

↓FA mobilization



↓TG

↓VLDL



↓LDL

↓HDL breakdown



↑HDL

Niacin

- Major side effects is **flushing**
 - Stimulates release of prostaglandins in skin
 - **Face** turns red, warm
 - Can blunt with **aspirin** (inhibits prostaglandin) prior to Niacin
 - Fades with time



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Niacin

- Hyperglycemia
 - Insulin resistance (mechanism incompletely understood)
 - Avoid in diabetes
- Hyperuricemia



Victor/Flickr



James Heilman, MD/Wikipedia

Fibrates

Gemfibrozil, clofibrate, bezafibrate, fenofibrate

- Activate **PPAR-a**
 - Modifies gene transcription
 - ↑ activity lipoprotein lipase
 - ↑ liver fatty acid oxidation → ↓ VLDL
- Major overall effect → TG breakdown
- Used for patients with very **high triglycerides**

Fibrates

Gemfibrozil, clofibrate, bezafibrate, fenofibrate

- Side effects
 - Myositis (Rhabdo with gemfibrozil; caution with statins)
 - ↑LFTs
 - Cholesterol gallstones

Absorption blockers

Ezetimibe

- Blocks cholesterol absorption
- Works at **intestinal brush border**
- Blocks dietary **cholesterol** absorption
 - Highly selective for cholesterol
 - Does not affect fat-soluble vitamins, triglycerides

Absorption blockers

Ezetimibe

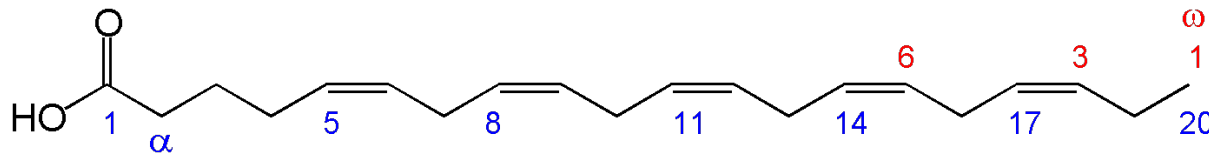
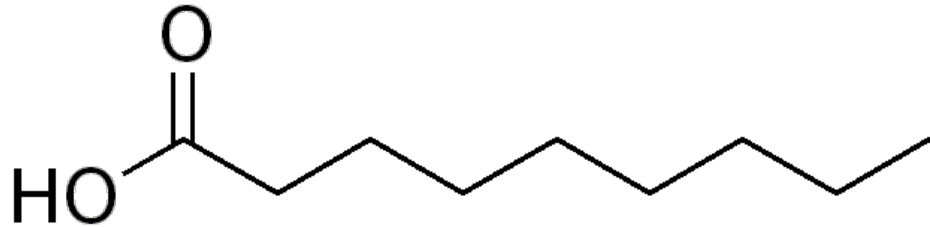
- Result: ↑LDL receptors on liver
- Modest reduction LDL
- Some ↓TG, ↑HDL
- Weak data on hard outcomes (MI, death)
- ↑LFTs
- Diarrhea

Bile Acid Resins

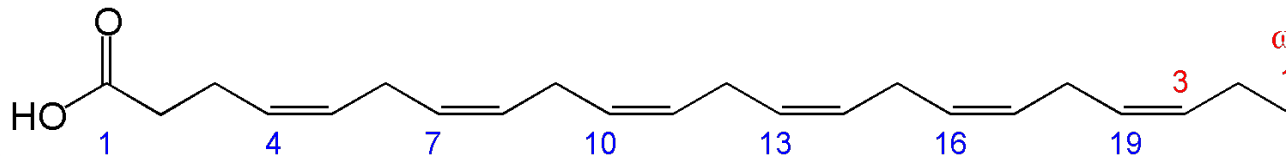
Cholestyramine, colestipol, colesevelam

- Old drugs; rarely used
- Prevent intestinal reabsorption bile
 - Cholesterol → bile → GI tract → reabsorption
- Resins lead to more bile excretion in stool
- Liver converts cholesterol → bile to makeup losses
- Modest lowering LDL
- Miserable for patients: **Bloating, bad taste**
- Can't absorb certain fat soluble vitamins
- Cholesterol gallstones

Omega-3 Fatty Acids



eicosapentaenoic acid (EPA)



docosahexaenoic acid (DHA)

Wikipedia/Public Domain

Omega-3 Fatty Acids

- Found in fish oil
- Consumption associated with ↓CV events
- Incorporated into cell membranes
- **Reduce VLDL production**
- **Lowers triglycerides** (~25 to 30%)
- Modest ↑ HDL
- Commercial supplements available (Lovaza)
- GI side effects: nausea, diarrhea, “fishy” taste

PCSK9 Inhibitors

Alirocumab, Evolocumab

- FDA approval in 2015
- PCSK9 → **degradation of LDL receptors**
 - Binds to LDL receptor
 - LDL receptor transported to lysosome
- Alirocumab/Evolocumab: Antibodies
- Inactivate PCSK9
 - ↓ LDL-receptor degradation
 - ↑ LDL receptors on hepatocytes
 - ↓ LDL cholesterol in plasma

PCSK9 Inhibitors

Alirocumab, Evolocumab

- Given by subcutaneous injection
- Results in significant LDL reductions ($>60\%$)
- Major adverse effect is injection site skin reaction
- Some association with memory problems